Reinforcement Learning for Adaptive Cancer Therapy

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***Abstract*—Adaptive cancer therapy proposes dynamic treatment strategies that aim to control tumor growth rather than eradicate the tumor completely. In this work we investigate whether reinforcement learning (RL) can automatically discover adaptive treatment policies. The tumor dynamics are modeled using a Lotka–Volterra competition model between sensitive and resistant cancer cells. An RL agent observes only the total tumor burden and its growth rate and learns treatment strategies that stabilize tumor size while minimizing therapy usage. We compare RL policies with classical treatment strategies including no treatment, maximum therapy, and threshold therapy. The results show that RL agents learn oscillatory treatment policies similar to adaptive therapy strategies used in oncology.**

***Index Terms*—reinforcement learning, adaptive therapy, cancer modeling, tumor dynamics**

I. Introduction

Traditional cancer therapy often applies continuous maximum tolerated dose (MTD). Although this may initially reduce tumor size, it frequently leads to the competitive release of resistant cancer cells.

Adaptive therapy proposes an alternative approach where treatment is applied dynamically depending on tumor evolution. Instead of eliminating all tumor cells, the objective is to maintain ecological competition between sensitive and resistant populations [5].

Mathematical models of tumor dynamics and treatment resistance have been widely studied in oncology [2]–[4]. The competition between sensitive and resistant populations can be naturally described using ecological models derived from classical population dynamics [1].

In this project we study whether reinforcement learning agents can discover such adaptive treatment strategies automatically from tumor observations.

The source code and implementation are available at **https://github.com/ZeevWeizmann/rl-cancer-therapy**. An interactive demonstration of the project is available at **https://zeevweizmann.github.io/rl-cancer-therapy/**.

II. Tumor Growth Model

Tumor dynamics are modeled as two competing populations:

- Sensitive cells T_s
- Resistant cells T_r

The dynamics follow a Lotka–Volterra competition model.

$$\frac{dT_s}{dt} = \alpha_s T_s \left(1 - \frac{T_s + c_{sr} T_r}{K}\right) - \beta u T_s \tag{1}$$

$$\frac{dT_r}{dt} = \alpha_r T_r \left(1 - \frac{T_r + c_{rs} T_s}{K}\right) \tag{2}$$

where

- K is the tumor carrying capacity
- α_s, α_r are growth rates
- c_{sr}, c_{rs} are competition coefficients
- u is treatment dose

Treatment affects only sensitive cells while resistant cells remain unaffected.

III. Reinforcement Learning Environment

The reinforcement learning agent does not observe the internal tumor populations directly.

Instead the observable state is

$$s = [\text{tumor size}, \text{growth rate}]$$

where tumor size corresponds to the normalized total population $T_s + T_r$.

A. Action Space

The agent selects among three therapy levels:

- 0.0 (no treatment)
- 0.5 (moderate therapy)
- 1.0 (maximum therapy)

B. Reward Function

The reward encourages tumor stabilization while penalizing excessive treatment:

$$R = -\left(\frac{T - T^*}{K}\right)^2 - 0.5 \max(dT, 0) - \lambda_d \cdot dose \tag{3}$$

where $T^* = 0.5K$ is the target tumor level.

IV. Algorithms

Two reinforcement learning algorithms were implemented using Stable-Baselines3:

- Deep Q-Network (DQN)
- Proximal Policy Optimization (PPO)

For comparison we also evaluate several baseline strategies:

- No treatment
- Maximum therapy
- Threshold adaptive therapy

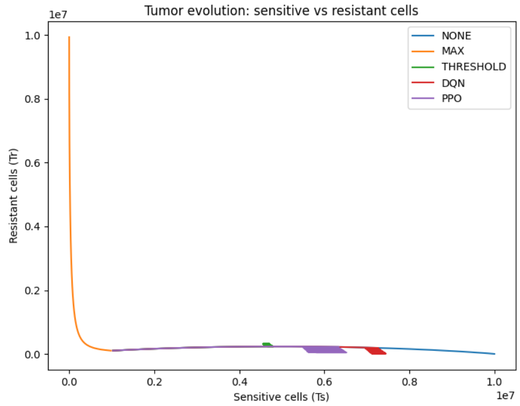

Fig. 1. Phase-space trajectories of tumor dynamics.

V. Results

Figure 1 shows phase-space trajectories of tumor populations.

Without treatment the tumor grows toward carrying capacity. Continuous therapy eliminates sensitive cells and leads to resistant dominance.

Adaptive strategies maintain competition between populations and stabilize tumor size.

Figure 2 shows the learning curves of RL agents.

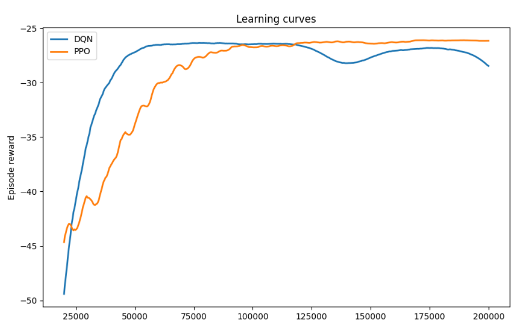

Fig. 2. Training performance of RL agents.

The PPO algorithm exhibits smoother convergence while DQN learns faster in early training.

Figure 3 shows the total tumor burden under different policies.

The RL agents learn oscillatory treatment cycles that maintain tumor size near the target level. PPO maintains a lower tumor burden compared to DQN, although the threshold policy achieves the lowest stable tumor size.

VI. Conclusion

This work demonstrates that reinforcement learning can automatically discover adaptive treatment strategies for tumor control.

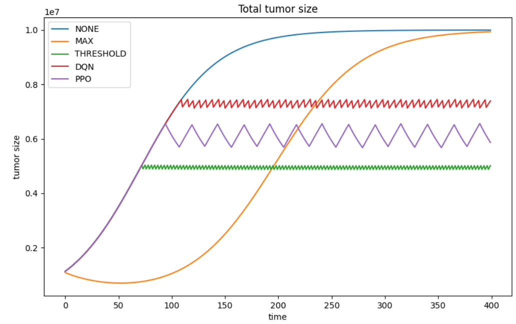

Fig. 3. Tumor burden under different treatment strategies.

The RL agents learn dynamic treatment cycles similar to threshold adaptive therapy strategies used in oncology.

Future work includes extending the model to more complex tumor dynamics and incorporating clinical data.

This work represents the first part of project. A second study will investigate reinforcement learning under more standard tumor growth models.

References

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